Unit 2 Pre-Test

Functions & Linear Functions

Standards assessed: NC.M1.F-IF.1 · F-IF.2 · F-IF.4 · F-IF.5 · F-IF.6 · F-IF.7 · A-CED.2 · F-LE.5

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.M1.F-IF.1 · NC.M1.F-IF.2 · NC.M1.F-IF.4 · NC.M1.F-IF.6 · NC.M1.A-CED.2 Recognize functions, use f(x) notation, and read lines.

• *I can tell if a relation is a function, evaluate f(x), read the slope and y-intercept, and write y = mx + b.*

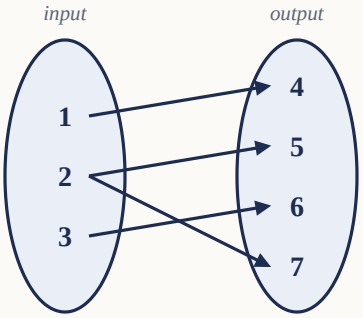
Answer each question. Use the picture to help you.

How to type your answers

number `-2` point `(0, 1)` equation `y=3x-2` interval `-3<=x<=4`

1. *Function?*

Is this relation a function? Write yes or no.



Answer: _____

2. *Evaluate*

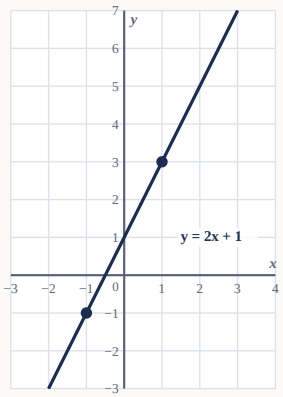
f(x) = 3x − 2. Find f(4).



f(4) = _____

3. *Key feature*

Use the graph. What is the **y-intercept**?



Answer: _____

4. *Rate of change*

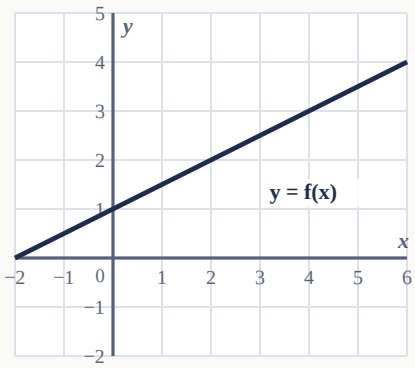
Use the table. What is the rate of change (slope)?

x	0	1	2	3
y	7	5	3	1

Answer: _____

5. *Read the graph*

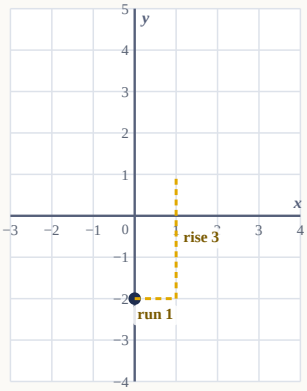
The graph shows y = f(x). Find f(4).



f(4) = _____

6. *Write the equation*

Write the equation of the line with slope 3 and y-intercept −2.



y = _____

SCORE **4.0** **Target Content**

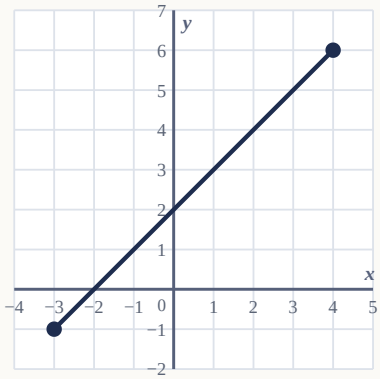
NC.M1.F-IF.5 · NC.M1.F-IF.6 · NC.M1.F-IF.7 · NC.M1.F-LE.5 · NC.M1.A-CED.2 Find domain, range, intercepts, and rates of change; write linear equations.

• *I can find domain and range, intercepts, and average rate of change, and write the equation of a line.*

Solve each problem. Show your work on paper.

7. *Domain*

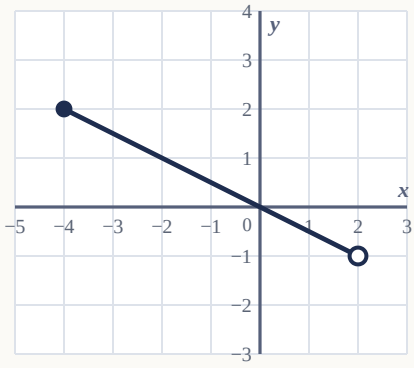
The graph has a closed dot at each end. What is the **domain**?



Answer: _____

8. *Range*

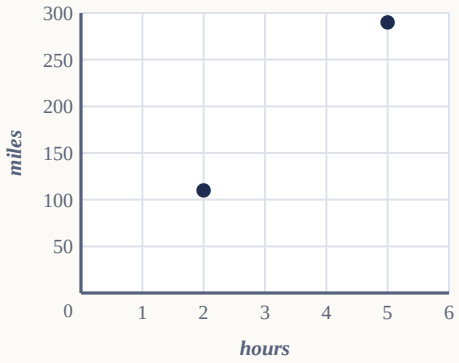
Closed dot on the left, open dot on the right. What is the **range**?



Answer: _____

9. *Average rate of change*

A car has gone 110 miles after 2 hours and 290 miles after 5 hours. Find the average rate of change in miles per hour.



rate = _____

10. *Intercepts*

Find the **x-intercept** of $2x + 3y = 12$.

x	0	**?**
y	**?**	0

Answer: _____

11. *Interpret*

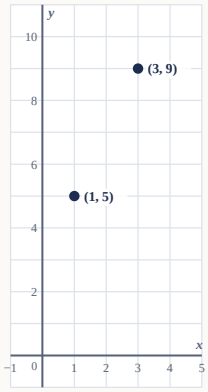
A taxi ride costs C = 2.5m + 40 dollars for m miles. What does the ride cost for **each mile**?

m	0	10	20
C	40	65	90

dollars = _____

12. *Write the equation*

Write the equation of the line through (1, 5) and (3, 9).



y = _____

SCORE **5.0** **Advanced Content**

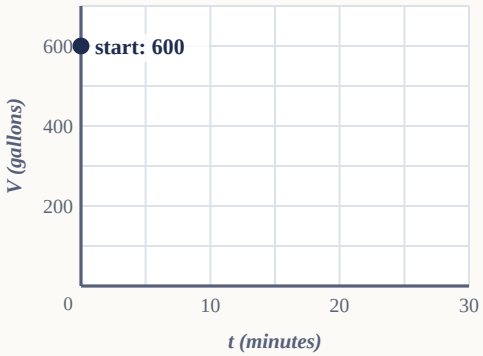
NC.M1.F-IF.2 · NC.M1.F-IF.4 · NC.M1.F-IF.5 · NC.M1.F-IF.6 · NC.M1.A-CED.2 · NC.M1.F-LE.5 Model with linear functions and explain what the numbers mean.

• *I can write a function for a situation, give a sensible domain, and explain how changing a number changes the graph.*

Solve each problem. Some answers are words.

13. *Write a function*

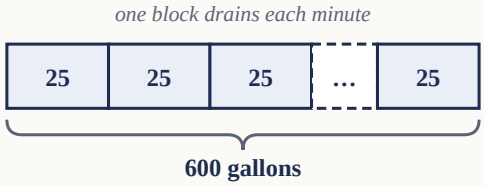
A pool holds 600 gallons. It drains 25 gallons each minute. Write V(t), the gallons left after t minutes.



V(t) = _____

14. *Domain in context*

For the pool, V(t) = 600 − 25t. What is a reasonable **domain**? (Time starts at 0 and stops when the pool is empty.)



Answer: _____

15. *Use function notation*

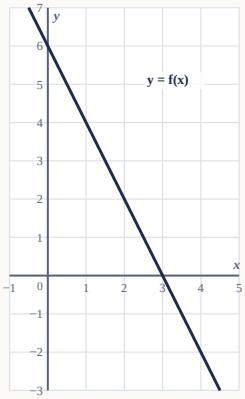
f(x) = 2x − 7. Find x when f(x) = 9.



x = _____

16. *Positive or negative*

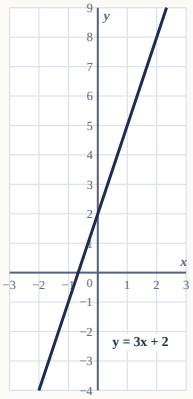
Use the graph of f. For which x-values is f(x) > 0?



Answer: _____

17. *Change a parameter*

f(x) = 3x + 2 is graphed. The 2 is changed to 7. Does the graph move **up or down**, and by how many units?



Answer: _____

18. *Average rate of change*

Find the average rate of change from x = 2 to x = 6.

x	0	2	4	6
y	3	4	8	15

Answer: _____